



Continuum Mechanics

Steven M. Wise

swise1@utk.edu
University of Tennessee

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Chapter 02

Tensor Calculus

Part 1



Fields and Order Symbols



Definition (2.1)

The standard Euclidean norms on $\mathbb{E}^3$, $\mathcal{V}$, and $\mathcal{V}^2$ are

$$|\mathbf{x}| = \sqrt{\mathbf{x} \cdot \mathbf{x}}, \quad \mathbf{x} \in \mathbb{E}^3,$$

$$|\mathbf{v}| = \sqrt{\mathbf{v} \cdot \mathbf{v}}, \quad \mathbf{v} \in \mathcal{V},$$

$$|S| = \sqrt{S : S}, \quad S \in \mathcal{V}^2.$$



Definition (2.2)

Consider a function

$$f : \mathcal{V} \longrightarrow W,$$

where W denotes any of the spaces $\mathbb{E}^3$, $\mathcal{V}$, $\mathcal{V}^2$, or $\mathbb{R}$.

Fix $r > 0$. We write

$$f(\mathbf{u}) = O(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0},$$

if there are constants $C, \delta > 0$ such that

$$0 < |\mathbf{u}| < \delta \implies |f(\mathbf{u})| \leq C|\mathbf{u}|^r.$$



Definition (2.2, Cont.)

For the fixed exponent $r > 0$, we write

$$f(\mathbf{u}) = o(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0},$$

if

$$\frac{|f(\mathbf{u})|}{|\mathbf{u}|^r} \rightarrow 0 \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

The symbols O and o are called *order symbols*.



Proposition

If

$$f(\mathbf{u}) = o(|\mathbf{u}|^r) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0},$$

then

$$f(\mathbf{u}) = O(|\mathbf{u}|^r) \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$



Proof.

Suppose that

$$f(\mathbf{u}) = o(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

Taking tolerance 1 in the limit definition of little o , there is a $\delta > 0$ such that

$$0 < |\mathbf{u}| < \delta \implies \frac{|f(\mathbf{u})|}{|\mathbf{u}|^r} < 1.$$

Thus, for $0 < |\mathbf{u}| < \delta$,

$$|f(\mathbf{u})| < |\mathbf{u}|^r.$$

Therefore $f(\mathbf{u}) = O(|\mathbf{u}|^r)$ as $\mathbf{u} \rightarrow \mathbf{0}$, with $C = 1$. □



Proposition

If

$$f(\mathbf{u}) = O(|\mathbf{u}|^p), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0},$$

and $p > r > 0$, *then*

$$f(\mathbf{u}) = o(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

Proof.

Exercise.



Remark

The statement

$$f(\mathbf{u}) = o(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0},$$

means that $|f(\mathbf{u})|$ *goes to zero faster than* $|\mathbf{u}|^r$.

Asymptotic Equality



Definition

Given two functions $f, g : \mathcal{V} \rightarrow W$, the notation

$$f(\mathbf{u}) = g(\mathbf{u}) + O(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0},$$

means

$$f(\mathbf{u}) - g(\mathbf{u}) = O(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

Similarly,

$$f(\mathbf{u}) = g(\mathbf{u}) + o(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0},$$

means

$$f(\mathbf{u}) - g(\mathbf{u}) = o(|\mathbf{u}|^r), \quad \text{as } \mathbf{u} \rightarrow \mathbf{0}.$$

Scalar, Vector, and Tensor Fields



Definition

Let $\Omega \subseteq \mathbb{E}^3$ be open. A function

$$\phi : \Omega \longrightarrow \mathbb{R}$$

is called a *scalar field*.

A function

$$\mathbf{q} : \Omega \longrightarrow \mathcal{V}$$

is called a *vector field*.

A function

$$S : \Omega \longrightarrow \mathcal{V}^2$$

is called a *tensor field*.



Differentiation of Scalar Fields



Definition (2.3)

Let $\Omega \subseteq \mathbb{R}^3$ be open. A scalar field $\phi : \Omega \rightarrow \mathbb{R}$ is said to be *differentiable* at $\mathbf{x} \in \Omega$ if there exists a vector $\nabla\phi(\mathbf{x}) \in \mathcal{V}$ such that

$$\phi(\mathbf{x} + \mathbf{h}) = \phi(\mathbf{x}) + \nabla\phi(\mathbf{x}) \cdot \mathbf{h} + o(|\mathbf{h}|), \quad \text{as } \mathbf{h} \rightarrow \mathbf{0},$$

with $\mathbf{x} + \mathbf{h} \in \Omega$. The vector $\nabla\phi(\mathbf{x})$ is called the *gradient* of ϕ at $\mathbf{x}$.

If ϕ is differentiable at $\mathbf{x}$, then, for every $\mathbf{a} \in \mathcal{V}$,

$$\nabla\phi(\mathbf{x}) \cdot \mathbf{a} = \left. \frac{d}{ds} \phi(\mathbf{x} + s\mathbf{a}) \right|_{s=0}, \quad s \in \mathbb{R}.$$

The Gradient of a Scalar Field



The directional characterization says that the scalar

$$\nabla\phi(\mathbf{x}) \cdot \mathbf{a}$$

is the derivative of ϕ at $\mathbf{x}$ in the direction $\mathbf{a}$.

The first-order approximation is

$$\phi(\mathbf{x} + \mathbf{h}) = \phi(\mathbf{x}) + \nabla\phi(\mathbf{x}) \cdot \mathbf{h} + o(|\mathbf{h}|), \quad \text{as } \mathbf{h} \rightarrow \mathbf{0},$$

with $\mathbf{x} + \mathbf{h} \in \Omega$.



Proposition

Let $\phi : \Omega \rightarrow \mathbb{R}$ be differentiable at $\mathbf{x} \in \Omega$. Then $\nabla\phi(\mathbf{x})$ is unique and satisfies the directional-derivative characterization given after Definition 2.3.

Proof.

See any good book on advanced calculus.





Result 2.4

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame. If

$$\mathbf{x} = x_i \mathbf{e}_i,$$

then

$$\nabla \phi(\mathbf{x}) = \frac{\partial \phi}{\partial x_i}(\mathbf{x}) \mathbf{e}_i.$$



Proof.

Let $\mathbf{a} \in \mathcal{V}$ be arbitrary and write

$$\mathbf{a} = a_i \mathbf{e}_i.$$

For any $s \in \mathbb{R}$,

$$\phi(\mathbf{x} + s\mathbf{a}) = \phi(x_1 + sa_1, x_2 + sa_2, x_3 + sa_3).$$

Using the chain rule,

$$\begin{aligned} \nabla \phi(\mathbf{x}) \cdot \mathbf{a} &= \left. \frac{d}{ds} \phi(\mathbf{x} + s\mathbf{a}) \right|_{s=0} \\ &= \left[\frac{\partial \phi}{\partial x_1}(\mathbf{x} + s\mathbf{a}) a_1 + \frac{\partial \phi}{\partial x_2}(\mathbf{x} + s\mathbf{a}) a_2 + \frac{\partial \phi}{\partial x_3}(\mathbf{x} + s\mathbf{a}) a_3 \right]_{s=0}. \end{aligned}$$



Proof (Cont.)

Therefore,

$$\begin{aligned}\nabla\phi(\mathbf{x}) \cdot \mathbf{a} &= \frac{\partial\phi}{\partial x_i}(\mathbf{x})a_i \\ &= \left(\frac{\partial\phi}{\partial x_j}(\mathbf{x})\mathbf{e}_j \right) \cdot \mathbf{a}.\end{aligned}$$

Since $\mathbf{a} \in \mathcal{V}$ is arbitrary,

$$\nabla\phi(\mathbf{x}) = \frac{\partial\phi}{\partial x_j}(\mathbf{x})\mathbf{e}_j.$$





Remark

The derivative of a scalar field $\phi : \Omega \rightarrow \mathbb{R}$ is the vector field

$$\nabla \phi : \Omega \longrightarrow \mathcal{V}.$$

Remark

We will often use the notation

$$\phi_{,i} := \frac{\partial \phi}{\partial x_i}.$$

Thus Result 2.4 may be written

$$\nabla \phi = \phi_{,i} \mathbf{e}_i.$$

Here i is a dummy index.



Proposition

Suppose that $\Omega \subset \mathbb{E}^3$ is bounded, open, and convex, and that $\phi \in C^1(\Omega; \mathbb{R})$. Assume there is a constant $\gamma > 0$ such that

$$|\nabla\phi(\mathbf{x}) - \nabla\phi(\mathbf{y})| \leq \gamma|\mathbf{x} - \mathbf{y}|$$

for all $\mathbf{x}, \mathbf{y} \in \Omega$. Then

$$|\phi(\mathbf{x}) - \phi(\mathbf{y}) - \nabla\phi(\mathbf{y}) \cdot (\mathbf{x} - \mathbf{y})| \leq \frac{\gamma}{2}|\mathbf{x} - \mathbf{y}|^2$$

for all $\mathbf{x}, \mathbf{y} \in \Omega$.

Proof.

See a good book on advanced calculus.



First-Order Approximation with Quadratic Remainder



Set

$$\mathbf{h} = \mathbf{x} - \mathbf{y} \in \mathcal{V}, \quad \mathbf{x} = \mathbf{y} + \mathbf{h}.$$

Then

$$|\phi(\mathbf{y} + \mathbf{h}) - \phi(\mathbf{y}) - \nabla\phi(\mathbf{y}) \cdot \mathbf{h}| \leq \frac{\gamma}{2} |\mathbf{h}|^2.$$

Hence

$$\phi(\mathbf{y} + \mathbf{h}) = \phi(\mathbf{y}) + \nabla\phi(\mathbf{y}) \cdot \mathbf{h} + O(|\mathbf{h}|^2), \quad \text{as } \mathbf{h} \rightarrow \mathbf{0},$$

Equivalently, replacing $\mathbf{y}$ by $\mathbf{x}$,

$$\phi(\mathbf{x} + \mathbf{h}) = \phi(\mathbf{x}) + \nabla\phi(\mathbf{x}) \cdot \mathbf{h} + O(|\mathbf{h}|^2), \quad \text{as } \mathbf{h} \rightarrow \mathbf{0},$$

where the increments are restricted by $\mathbf{y} + \mathbf{h} \in \Omega$ or $\mathbf{x} + \mathbf{h} \in \Omega$, respectively.



Differentiation of Vector Fields



Definition (2.5)

Let $\Omega \subseteq \mathbb{R}^3$ be open. A vector field $\mathbf{v} : \Omega \rightarrow \mathcal{V}$ is said to be *differentiable* at $\mathbf{x} \in \Omega$ if there exists a second-order tensor $\nabla \mathbf{v}(\mathbf{x}) \in \mathcal{V}^2$ such that

$$\mathbf{v}(\mathbf{x} + \mathbf{h}) = \mathbf{v}(\mathbf{x}) + \nabla \mathbf{v}(\mathbf{x})\mathbf{h} + o(|\mathbf{h}|), \quad \text{as } \mathbf{h} \rightarrow \mathbf{0},$$

with $\mathbf{x} + \mathbf{h} \in \Omega$. The tensor $\nabla \mathbf{v}(\mathbf{x})$ is called the *gradient* of $\mathbf{v}$ at $\mathbf{x}$.

If $\mathbf{v}$ is differentiable at $\mathbf{x}$, then, for every $\mathbf{a} \in \mathcal{V}$,

$$\nabla \mathbf{v}(\mathbf{x})\mathbf{a} = \left. \frac{d}{d\alpha} \mathbf{v}(\mathbf{x} + \alpha \mathbf{a}) \right|_{\alpha=0}, \quad \alpha \in \mathbb{R}.$$



Result 2.6

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame, and suppose that

$$\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i, \quad \mathbf{x} = x_i\mathbf{e}_i.$$

Then

$$\nabla \mathbf{v}(\mathbf{x}) = \frac{\partial v_i}{\partial x_j}(\mathbf{x})\mathbf{e}_i \otimes \mathbf{e}_j.$$



Proof.

Let $\mathbf{a} = a_k \mathbf{e}_k \in \mathcal{V}$ be arbitrary. For each component,

$$v_i(\mathbf{x} + \alpha \mathbf{a}) = v_i(x_1 + \alpha a_1, x_2 + \alpha a_2, x_3 + \alpha a_3).$$

By the chain rule,

$$\begin{aligned} \left. \frac{d}{d\alpha} v_i(\mathbf{x} + \alpha \mathbf{a}) \right|_{\alpha=0} &= \left[\frac{\partial v_i}{\partial x_1}(\mathbf{x} + \alpha \mathbf{a}) a_1 + \frac{\partial v_i}{\partial x_2}(\mathbf{x} + \alpha \mathbf{a}) a_2 \right. \\ &\quad \left. + \frac{\partial v_i}{\partial x_3}(\mathbf{x} + \alpha \mathbf{a}) a_3 \right]_{\alpha=0} \\ &= \frac{\partial v_i}{\partial x_j}(\mathbf{x}) a_j. \end{aligned}$$



Proof (Cont.)

Therefore,

$$\begin{aligned}\nabla \mathbf{v}(\mathbf{x})\mathbf{a} &= \left. \frac{d}{d\alpha} \mathbf{v}(\mathbf{x} + \alpha \mathbf{a}) \right|_{\alpha=0} \\ &= \left. \frac{d}{d\alpha} v_i(\mathbf{x} + \alpha \mathbf{a}) \right|_{\alpha=0} \mathbf{e}_i \\ &= \frac{\partial v_i}{\partial x_j}(\mathbf{x}) a_j \mathbf{e}_i \\ &= \left(\frac{\partial v_i}{\partial x_j}(\mathbf{x}) \mathbf{e}_i \otimes \mathbf{e}_j \right) \mathbf{a}.\end{aligned}$$

Since $\mathbf{a}$ is arbitrary,

$$\nabla \mathbf{v}(\mathbf{x}) = \frac{\partial v_i}{\partial x_j}(\mathbf{x}) \mathbf{e}_i \otimes \mathbf{e}_j.$$





Remark

The derivative of a vector field $\mathbf{v} : \Omega \rightarrow \mathcal{V}$ is the second-order tensor field

$$\nabla \mathbf{v} : \Omega \longrightarrow \mathcal{V}^2.$$

Remark

As for scalar fields, we often write

$$v_{i,j} := \frac{\partial v_i}{\partial x_j}.$$

Thus

$$\nabla \mathbf{v} = v_{i,j} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Here i, j are dummy indices.



Proposition

Suppose that $\Omega \subset \mathbb{E}^3$ is bounded, open, and convex, and that $\mathbf{v} \in C^1(\Omega; \mathcal{V})$. Assume there is a constant $\gamma > 0$ such that

$$|\nabla \mathbf{v}(\mathbf{x}) - \nabla \mathbf{v}(\mathbf{y})| \leq \gamma |\mathbf{x} - \mathbf{y}|$$

for all $\mathbf{x}, \mathbf{y} \in \Omega$. Then

$$|\mathbf{v}(\mathbf{x}) - \mathbf{v}(\mathbf{y}) - \nabla \mathbf{v}(\mathbf{y})(\mathbf{x} - \mathbf{y})| \leq \frac{\gamma}{2} |\mathbf{x} - \mathbf{y}|^2$$

for all $\mathbf{x}, \mathbf{y} \in \Omega$.

Proof.

See a good book on advanced calculus.



First-Order Approximation with Quadratic Remainder



As before, set $\mathbf{h} = \mathbf{x} - \mathbf{y}$. Then

$$\mathbf{v}(\mathbf{y} + \mathbf{h}) = \mathbf{v}(\mathbf{y}) + \nabla \mathbf{v}(\mathbf{y})\mathbf{h} + O(|\mathbf{h}|^2), \quad \text{as } \mathbf{h} \rightarrow \mathbf{0},$$

Equivalently,

$$\mathbf{v}(\mathbf{x} + \mathbf{h}) = \mathbf{v}(\mathbf{x}) + \nabla \mathbf{v}(\mathbf{x})\mathbf{h} + O(|\mathbf{h}|^2), \quad \text{as } \mathbf{h} \rightarrow \mathbf{0},$$

where the increments are restricted by $\mathbf{y} + \mathbf{h} \in \Omega$ or $\mathbf{x} + \mathbf{h} \in \Omega$, respectively.



Divergence and Product Rules



Definition (2.7)

Let $\Omega \subseteq \mathbb{E}^3$ be open. To every differentiable vector field $\mathbf{v} : \Omega \rightarrow \mathcal{V}$, we associate the scalar field

$$\nabla \cdot \mathbf{v} : \Omega \longrightarrow \mathbb{R}$$

defined by

$$\nabla \cdot \mathbf{v} = \text{tr}(\nabla \mathbf{v}).$$

The scalar field $\nabla \cdot \mathbf{v}$ is called the *divergence* of $\mathbf{v}$.



Result 2.8

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame. Write

$$\mathbf{v}(\mathbf{x}) = v_i(\mathbf{x})\mathbf{e}_i,$$

and assume that $\mathbf{v} : \Omega \rightarrow \mathcal{V}$ is differentiable. Then

$$\nabla \cdot \mathbf{v}(\mathbf{x}) = \frac{\partial v_i}{\partial x_i}(\mathbf{x}) = v_{i,i}(\mathbf{x}).$$

The repeated index i is a dummy index.

Proof.

Exercise. ☐



Definition (2.9)

Let $\Omega \subseteq \mathbb{E}^3$ be open. To every differentiable second-order tensor field

$$S : \Omega \longrightarrow \mathcal{V}^2,$$

we associate the vector field

$$\nabla \cdot S : \Omega \longrightarrow \mathcal{V}$$

defined by the relation

$$(\nabla \cdot S) \cdot \mathbf{a} = \nabla \cdot (S^T \mathbf{a})$$

for every constant vector $\mathbf{a} \in \mathcal{V}$.

We call $\nabla \cdot S$ the *divergence* of S .



Result 2.10

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame and assume that $S : \Omega \rightarrow \mathcal{V}^2$ is differentiable. If

$$S = S_{ij}\mathbf{e}_i \otimes \mathbf{e}_j,$$

then

$$(\nabla \cdot S)(\mathbf{x}) = \frac{\partial S_{ij}}{\partial x_j}(\mathbf{x})\mathbf{e}_i = S_{ij,j}(\mathbf{x})\mathbf{e}_i.$$

Equivalently,

$$[\nabla \cdot S]_i = S_{ij,j}.$$

In this component identity i is free and j is dummy; in the vector expansion above, both indices are summed.



Proof.

Let $\mathbf{a} = a_k \mathbf{e}_k \in \mathcal{V}$ be an arbitrary constant vector and set

$$\mathbf{q} := S^T \mathbf{a}.$$

Write $\mathbf{q} = q_j \mathbf{e}_j$. Since

$$S = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j,$$

we have

$$q_j = [\mathbf{q}]_j = S_{ij} a_i.$$

Using Definition 2.9 and Result 2.8,

$$\begin{aligned} (\nabla \cdot S) \cdot \mathbf{a} &= \nabla \cdot (S^T \mathbf{a}) \\ &= \nabla \cdot \mathbf{q} \\ &= q_{j,j}. \end{aligned}$$



Proof (Cont.)

Because $\mathbf{a}$ is constant,

$$\begin{aligned}(\nabla \cdot S) \cdot \mathbf{a} &= \frac{\partial q_j}{\partial x_j} \\&= \frac{\partial}{\partial x_j} (S_{ij} a_i) \\&= \frac{\partial S_{ij}}{\partial x_j} a_i \\&= S_{ij,j} a_i \\&= (S_{ij,j} \mathbf{e}_i) \cdot \mathbf{a}.\end{aligned}$$

Since $\mathbf{a} \in \mathcal{V}$ is arbitrary,

$$\nabla \cdot S = S_{ij,j} \mathbf{e}_i.$$





Result 2.11 (Product Rules)

Let ϕ , $\mathbf{v}$, and S be differentiable scalar, vector, and tensor fields, respectively. Then

$$\nabla(\phi\mathbf{v}) = \mathbf{v} \otimes \nabla\phi + \phi\nabla\mathbf{v},$$

$$\nabla \cdot (\phi S) = \phi \nabla \cdot S + S \nabla \phi,$$

$$\nabla \cdot (S^T \mathbf{v}) = (\nabla \cdot S) \cdot \mathbf{v} + S : \nabla \mathbf{v}.$$



Proof.

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame and write

$$\mathbf{v} = v_i \mathbf{e}_i, \quad S = S_{ij} \mathbf{e}_i \otimes \mathbf{e}_j.$$

Using Result 2.6,

$$\begin{aligned} \nabla(\phi \mathbf{v}) &= \frac{\partial(\phi v_i)}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= \frac{\partial \phi}{\partial x_j} v_i \mathbf{e}_i \otimes \mathbf{e}_j + \phi \frac{\partial v_i}{\partial x_j} \mathbf{e}_i \otimes \mathbf{e}_j \\ &= (v_i \mathbf{e}_i) \otimes \left(\frac{\partial \phi}{\partial x_j} \mathbf{e}_j \right) + \phi \nabla \mathbf{v} \\ &= \mathbf{v} \otimes \nabla \phi + \phi \nabla \mathbf{v}. \end{aligned}$$



Proof (Cont.)

Next, using Result 2.10,

$$\begin{aligned}\nabla \cdot (\phi S) &= \frac{\partial(\phi S_{ij})}{\partial x_j} \mathbf{e}_i \\ &= \left(\frac{\partial \phi}{\partial x_j} S_{ij} + \phi \frac{\partial S_{ij}}{\partial x_j} \right) \mathbf{e}_i \\ &= S_{ij} \frac{\partial \phi}{\partial x_j} \mathbf{e}_i + \phi \nabla \cdot S \\ &= S \nabla \phi + \phi \nabla \cdot S.\end{aligned}$$



Proof (Cont.)

Finally, $S^T \mathbf{v}$ is a vector field. In the frame $\mathcal{F}$,

$$\begin{aligned} S^T \mathbf{v} &= [S^T]_{ij} v_j \mathbf{e}_i \\ &= S_{ji} v_j \mathbf{e}_i \\ &= S_{ij} v_i \mathbf{e}_j, \end{aligned}$$

where the last equality consistently renames the dummy indices.

Using Result 2.8,

$$\begin{aligned} \nabla \cdot (S^T \mathbf{v}) &= \frac{\partial (S_{ij} v_i)}{\partial x_j} \\ &= \frac{\partial S_{ij}}{\partial x_j} v_i + S_{ij} \frac{\partial v_i}{\partial x_j}. \end{aligned}$$



Proof (Cont.)

The first term is

$$\frac{\partial S_{ij}}{\partial x_j} v_i = (\nabla \cdot S) \cdot \mathbf{v}.$$

The second term is

$$S_{ij} \frac{\partial v_i}{\partial x_j} = S_{ij} [\nabla \mathbf{v}]_{ij} = S : \nabla \mathbf{v}.$$

Therefore,

$$\nabla \cdot (S^T \mathbf{v}) = (\nabla \cdot S) \cdot \mathbf{v} + S : \nabla \mathbf{v}.$$





Worked Exercises



Example (Exercise 2.4)

Let $\Omega \subseteq \mathbb{R}^3$ be open, and let

$$\phi : \Omega \rightarrow \mathbb{R}, \quad \mathbf{v}, \mathbf{w} : \Omega \rightarrow \mathcal{V}, \quad S : \Omega \rightarrow \mathcal{V}^2$$

be differentiable fields. Show that

(a)

$$\nabla \cdot (\phi \mathbf{v}) = \nabla \phi \cdot \mathbf{v} + \phi \nabla \cdot \mathbf{v};$$

(b)

$$\nabla(\mathbf{v} \cdot \mathbf{w}) = (\nabla \mathbf{v})^T \mathbf{w} + (\nabla \mathbf{w})^T \mathbf{v};$$

(c)

$$\nabla \cdot (\mathbf{v} \otimes \mathbf{w}) = (\nabla \mathbf{v}) \mathbf{w} + (\nabla \cdot \mathbf{w}) \mathbf{v}.$$



Example (Exercise 2.4(a))

Using Result 2.8 and the scalar product rule,

$$\begin{aligned}\nabla \cdot (\phi \mathbf{v}) &= \frac{\partial}{\partial x_i} (\phi v_i) \\ &= \frac{\partial \phi}{\partial x_i} v_i + \phi \frac{\partial v_i}{\partial x_i} \\ &= \nabla \phi \cdot \mathbf{v} + \phi (\nabla \cdot \mathbf{v}).\end{aligned}$$

Here i is a dummy index in every term; there is no free index because the result is a scalar.



Example (Exercise 2.4(b))

Let $\mathcal{F} = \{\mathbf{e}_i\}$ be an orthonormal frame. Since

$$\mathbf{v} \cdot \mathbf{w} = v_k w_k,$$

we have

$$\begin{aligned}\nabla(\mathbf{v} \cdot \mathbf{w}) &= \frac{\partial}{\partial x_i} (v_k w_k) \mathbf{e}_i \\ &= \left(\frac{\partial v_k}{\partial x_i} w_k + v_k \frac{\partial w_k}{\partial x_i} \right) \mathbf{e}_i \\ &= \left([(\nabla \mathbf{v})^T]_{ik} w_k + [(\nabla \mathbf{w})^T]_{ik} v_k \right) \mathbf{e}_i.\end{aligned}$$

Here i is free in the component identity and k is dummy.



Example (Exercise 2.4(b), Cont.)

Using square brackets for component extraction,

$$\left[(\nabla \mathbf{v})^T \mathbf{w} \right]_i = [(\nabla \mathbf{v})^T]_{ik} w_k,$$

$$\left[(\nabla \mathbf{w})^T \mathbf{v} \right]_i = [(\nabla \mathbf{w})^T]_{ik} v_k.$$

Therefore,

$$\nabla(\mathbf{v} \cdot \mathbf{w}) = (\nabla \mathbf{v})^T \mathbf{w} + (\nabla \mathbf{w})^T \mathbf{v}.$$

Both sides have the same free component index i ; k is dummy.



Example (Exercise 2.4(c))

The components of the dyad $\mathbf{v} \otimes \mathbf{w}$ are

$$[\mathbf{v} \otimes \mathbf{w}]_{ij} = v_i w_j.$$

Thus, by Result 2.10,

$$\begin{aligned}\nabla \cdot (\mathbf{v} \otimes \mathbf{w}) &= \frac{\partial}{\partial x_j} (v_i w_j) \mathbf{e}_i \\ &= \left(\frac{\partial v_i}{\partial x_j} w_j + v_i \frac{\partial w_j}{\partial x_j} \right) \mathbf{e}_i.\end{aligned}$$

Here i is free in the component identity and j is dummy.



Example (Exercise 2.4(c), Cont.)

The first term satisfies

$$\frac{\partial v_i}{\partial x_j} w_j = [(\nabla \mathbf{v}) \mathbf{w}]_i,$$

and the second satisfies

$$v_i \frac{\partial w_j}{\partial x_j} = [(\nabla \cdot \mathbf{w}) \mathbf{v}]_i.$$

Therefore,

$$\nabla \cdot (\mathbf{v} \otimes \mathbf{w}) = (\nabla \mathbf{v}) \mathbf{w} + (\nabla \cdot \mathbf{w}) \mathbf{v}.$$

Again, i is free and j is dummy in the component calculation.